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FACULTY OF APPLIED SCIENCES

IT & BUSINESS ANALYTICS PROGRAM

Key factors and trends of Ukrainian IT sector

Second interim report

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1 Topic and motivation

The focus of this research is to analyze a variety of factors that may be related to the income earned by IT professionals in Ukraine. The ultimate goal is to evaluate all available correlations and select some of the most significant ones. With the rapid growth of the IT sector in Ukraine, it may be very useful for businesses, specifically within the technological industry, their employees, both governmental and non-governmental organizations dealing with the economic implications of the IT trends.

Motivation:

Although the IT industry has been growing consistently for the last decades, there are a variety of underlying trends within this job market. Employers, job seekers, and educational institutions can benefit from insights into salary trends, helping them make informed decisions. Understanding the major determinants of salary allows firms to optimize compensation strategies to attract and retain talent. This research is relevant to HR professionals, tech companies, recruitment agencies, and policymakers seeking to understand the dynamics of the labor market.

2 Aim and Tasks

Research Aim

The primary goal of this research is to identify and quantify the significant factors that influence the salaries of IT specialists in Ukraine.

Research Questions

- What are the main factors affecting IT salaries in Ukraine?
- How do regional differences impact salaries?
- What is the effect of English proficiency on earnings?

- How do salary trends evolve over time?
- What is the wage disparity between different programming languages within IT?
- Is there a sizable wage gap between genders?

Tasks

- Collect and clean the dataset to ensure consistency.
- Perform exploratory data analysis to identify trends and distributions.
- Apply econometric models to estimate the impact of various factors on salaries.
- Compare findings with existing literature.

3 Literature Review

Our study builds on previous research in labor economics, wage determinants, and IT industry salary studies.

1. "Bargaining, Sorting, and the Gender Wage Gap", Card, Cardoso, and Kline (2016). It provides a basis for understanding wage inequality within companies. While the focus is on gender disparities, their model of wage bargaining within firms is applicable to broader wage inequality analyses, including factors such as skills and experience.
2. "Macroeconomic Forecasting using Dynamic Factor Models", Stock and Watson (2023). This article introduces advanced time-series techniques that are highly applicable for forecasting market trends, including wages. Their work on dynamic factor models and their application to the labor market data provides a basis for analyzing salary trends over time.

3. "The Growing Importance of Social Skills in the Labor Market", Deming and Noray (2017). In this paper, authors talk about the increasing demand for both technical and soft skills in the technology sector. Their research shows that while programming expertise is crucial, higher wages are awarded to IT professionals with a combination of technical and managerial skills.
4. "Introductory Econometrics: A Modern Approach", Wooldridge (2020). This textbook serves as a fundamental reference for the econometric methods used in our study. Wooldridge's work provides a comprehensive introduction to regression analysis, panel data techniques, and instrumental variable methods, all of which are essential for analyzing wage determinants and estimating causal effects in labor economics.

4 Data description

Our analysis is based on five datasets containing survey data on IT professionals in Ukraine. The datasets include variables such as:

- Demographics: Age, gender, location.
- Education: Degree type, university attended, scientific degree.
- Job Information: Position, specialization, programming languages, frameworks used, platforms developed for.
- Experience: Years in the IT industry, years at the current company, number of companies worked at.
- Employment Status: Currently working in IT, job location.
- Industry: Sector of employment, main company activity.
- Company Type: Startup vs. large firm, Ukrainian vs. foreign company, company size in Ukraine.
- Compensation: Monthly salary (USD), changes in salary over the last year, bonuses, hourly rate for clients.
- English Proficiency: Self-reported level of English knowledge.

Data Source: DOU research of IT salaries
The datasets were obtained from salary surveys conducted among IT professionals in Ukraine. We will ensure data quality by addressing missing values, inconsistencies, and potential biases in self-reported data.

4.1 Expected Challenges

- Self-Selection Bias: Respondents may not represent the entire IT workforce.
- Currency Fluctuations: Since salaries are in USD, exchange rate volatility may affect analysis.
- Non-Standardized Titles: Job titles and roles may vary across companies, requiring categorization.
- Overlapping Samples: Respondents may appear in multiple datasets, hampering time-series analysis.

4.2 EDA

We begin with the general salary landscape. Figure 1 illustrates the distribution of monthly salaries among IT professionals in Ukraine. The distribution is highly right-skewed, with most salaries falling within the \$500–\$5000 range.

English proficiency significantly impacts salary outcomes. Figure 2 demonstrates that professionals with Fluent or Advanced English skills consistently earn higher salaries with broader distributions, while those with Intermediate or lower proficiency show compressed salary ranges and lower medians. This pattern highlights the

importance of language skills as a gateway to higher-paying opportunities.

Gender also shapes salary outcomes. Figure 3 shows that male professionals' salary distribution tends to be slightly higher and more dispersed compared to females. Although both groups span similar ranges, males are overrepresented among top earners.

Experience emerges as a fundamental driver of salary growth. Figure 4 reveals a clear positive correlation between years of experience and salary, with rapid increases during early career years but plateauing after the 10-year mark, indicating diminishing returns, as shown in Figure 5.

Company type appears to influence compensation structures, with preliminary data (Figure 6) suggesting product companies offer higher average salaries than other firms. This supports our hypothesis that product companies, which often generate higher revenue per employee and retain greater control over intellectual prop-

erty, compensate their employees more generously.

Programming language specialization also correlates with salary differences, as seen in Figure 7. Our analysis indicates varying compensation levels across language ecosystems, potentially reflecting market demand and application complexity.

Despite the rise of remote work, regional differences persist in the Ukrainian IT market. Our data (Figure 8) reveals geographic variations in compensation that may reflect differences in local cost of living, concentration of employers, and regional economic development.

These exploratory insights support our six key hypotheses: that salary varies significantly between company types (H1), increases with English proficiency (H2), grows with total experience (H3), varies across programming languages (H4), reflects gender disparities (H5), and differs by region (H6). We now proceed to formal statistical testing of these hypotheses.

5 Methods

5.1 Ordinary Least Squares (OLS)

To analyze the main factors that have an effect on salaries, we will use the following multiple linear regression model:

$$\begin{aligned} \text{Salary}_i = & \beta_0 + \beta_1 \cdot \text{CompanyType}_i + \beta_2 \cdot \text{Experience}_i + \beta_3 \cdot \text{EnglishLevel}_i \\ & + \beta_4 \cdot \text{Male}_i + \beta_5 \cdot \text{FullTime}_i \\ & + \sum_j \gamma_j \cdot \text{Year}_{ij} + \sum_k \delta_k \cdot \text{Title}_{ik} + \sum_l \theta_l \cdot \text{Language}_{il} + \epsilon_i \end{aligned} \quad (1)$$

This specification was chosen based on model selection using the Akaike Information Criterion (AIC), which balances model fit and complexity. The final model includes only those variables that contributed significantly to explaining variation in salaries while minimizing AIC.

Dependent variable:

- Salary: Monthly salary in USD.

Independent variables:

- **CompanyType:** Dummy variable indicating whether the company is a product company.
- **Experience:** Total years of professional experience.
- **English Level:** Self-reported English proficiency level.
- **Male:** Dummy variable for gender (1 = male, 0 = female).
- **Full Time:** Dummy variable for full-time employment status.
- **Year:** Categorical variable representing the survey year.
- **Title:** Categorical variable for job title (e.g., Junior, Middle, Senior).
- **Programming Language:** Categorical variable for primary programming language used.

OLS regression will allow us to estimate how each of these factors contributes to salary variation, while controlling for fixed effects across year, job title, and programming language.

5.2 Log-Linear Regression

As shown in Figure 1 (left plot), the original salary distribution is highly right-skewed, with a long tail of high earners. This violates the OLS assumption of normally distributed residuals. Taking the logarithm of salary (right plot) makes the distribution approximately normal, which improves model fit.

$$\log(\text{Salary}_i) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \epsilon_i \quad (2)$$

5.3 Hierarchical Linear Model (HLM)

To account for unobserved heterogeneity across different regions of Ukraine, we estimate a hierarchical linear model (HLM) using a mixed effects approach. The model specification is as follows:

$$\begin{aligned} \log(\text{Salary})_{ij} = & \beta_0 + \beta_1 \cdot \text{Experience}_{ij} + \beta_2 \cdot \text{EnglishLevel}_{ij} + \beta_3 \cdot \text{Male}_{ij} + \beta_4 \cdot \text{FullTime}_{ij} \\ & + \sum_k \gamma_k \cdot \text{Title}_{ijk} + \sum_l \theta_l \cdot \text{Language}_{ijl} + \sum_m \delta_m \cdot \text{Year}_{ijm} \\ & + u_{0j} + u_{1j} \cdot \text{Year}_{ij} + \epsilon_{ij} \end{aligned} \quad (3)$$

where i indexes individuals and j indexes regions.

Dependent variable:

- *Log(Salary)*: Natural logarithm of monthly salary in USD.

Independent variables (fixed effects):

- **Experience:** Total years of professional experience.
- **English Level:** Self-reported English proficiency level.

- Male: Dummy variable for gender (1 = male, 0 = female).
- Full Time: Dummy variable for full-time employment status.
- Title: Categorical variable for job title (e.g., Junior, Middle, Senior).
- Programming Language: Categorical variable for primary programming language used.
- Year: Categorical variable representing the survey year.

Random effects:

- Random intercept (u_{0j}) at the region level, allowing baseline salaries to vary across regions.
- Random slope on Year (u_{1j}), allowing the effect of year (salary growth or decline over time) to vary across regions.

This model structure allows us to control for both individual-level characteristics and regional differences in salary levels and salary dynamics over time. By introducing random intercepts and random slopes, we can more accurately model the hierarchical structure of the data and reduce bias due to omitted regional factors. Moreover, the inclusion of a random slope on Year captures how different regions experience different rates of salary growth or decline, rather than assuming a uniform trend across the country.

5.4 Heteroskedasticity test

To assess whether the variance of residuals remains constant across all observations — a key assumption of OLS — we conducted the Breusch–Pagan test for heteroskedasticity. The test produced the following results:

- LM statistic: 1341.71
- LM-Test p-value: 6.87×10^{-217}
- F-statistic: 14.06
- F-Test p-value: 6.38×10^{-222}

These extremely low p-values strongly reject the null hypothesis of homoskedasticity, indicating that **heteroskedasticity is present** in the model. As a result, we estimate our regression using robust standard errors to ensure valid inference despite the presence of non-constant error variance.

5.5 Multicollinearity Testing

To check for multicollinearity, we calculated Variance Inflation Factors for all predictors

in our model. This diagnostic helps us identify if any variables are redundant or distort the stability of our coefficient estimates.

In general, most variables showed no signs of multicollinearity. However, some programming languages and job titles had moderately high Variance Inflation Factors values, which likely reflects real-world patterns — for example, certain roles being strongly tied to specific technologies or company types. These results are not unexpected in a dataset related to IT salaries and do not pose a threat to the model’s validity. We chose to keep all predictors in the model, as they remain meaningful and interpretable, and their inclusion improves the explanatory power of the regression.

6 Hypotheses

Our project investigates the key factors that influence IT salaries in Ukraine, a rapidly evolving and globally integrated job market. Based on the exploratory data analysis

and literature, we formulated the following hypotheses to test by using models. Each hypothesis reflects a core research question driven by economic theory, industry observations and patterns in our data.

Effect of company type

- *Research Question:* Do product companies pay more than outsourcing firms?
Product companies often generate higher revenues per employee and have greater control over their products and intellectual property. In contrast, outsourcing firms typically operate under tighter client budgets and are often forced to reduce labor costs. We believe that IT professionals working in product companies tend to earn higher salaries.
- Hypothesis: Product companies pay higher salaries than outsourcing companies.
- $H_0: \beta_{product} \leq 0$
 $H_1: \beta_{product} > 0$

Impact of English proficiency

- *Research Question:* Is a higher level of English associated with a higher salary?
English proficiency is essential in the Ukrainian IT market due to its strong international orientation. Most clients, documentation, and collaboration are in English. We test whether higher English proficiency leads to increased earnings, suggesting that communication skills open access to higher-paying opportunities.
- Hypothesis: Higher levels of English proficiency are associated with higher salaries
- H_0 : The level of English proficiency does not affect the average salary.

H_1 : As the level of English grows, the average salary increases.

- $H_0: \beta_{englishlevel} = 0$
 $H_1: \beta_{english\ level} > 0$

Impact of working experience

- *Research Question:* How much does experience matter for IT salaries?
The labor economics literature mentions the human capital theory, where accumulated experience increases productivity and therefore earnings. We examine whether years of total experience in the IT industry significantly affect the income.
- Hypothesis: Greater total experience leads to increased salary
- H_0 : Total experience does not affect the average salary
 H_1 : Salary increase with total experience
- $H_0: \beta_{total\ experience} = 0$
 $H_1: \beta_{total\ experience} > 0$

Effect of programming language

- *Research Question:* Do salaries vary by programming language used?
Different programming languages are in demand in different segments of the market. For example, developers using C++ or Rust may be concentrated in high-performance applications, while PHP developers may be working on legacy or lower-budget projects. We test if language choice is associated with different salary levels.
- Hypothesis: Programming language significantly affects the average salary.
- H_0 : The programming language does not affect the average salary.
 H_1 : The average salary differs significantly depending on the programming language.

- $H_0 : \beta_{\text{programming language}} = 0 \quad \forall k$
 $H_1: \exists k \text{ such that } \beta_{\text{programming language}}, k \neq 0$

Gender pay gap

- *Research Question:* Is there evidence of gender-based wage difference?
 While the IT industry promotes equality and skill-based salaries, global data often reveals persistent gender wage gaps. We explore whether gender has a statistically significant impact on salaries, controlling for other factors such as experience, position, and company type.
- Hypothesis: Gender has an effect on salary
- H_0 : Gender does not affect salary
 H_1 : Gender do affect salary
- $H_0: \beta_{\text{male}} = 0$
 $H_1: \beta_{\text{male}} \neq 0$

Regional salary difference

- *Research Question:* Does location still matter in a remote work world?
 With the rise of remote work, the physical location in determining salary may not be significant. Nonetheless, regional economic conditions, cost of living, and local labor market dynamics may still play a role. We test whether region still has a significant effect on earnings.
- Hypothesis: Region has a significant effect on salary
- H_0 : Region does not have a statistically significant effect on salary after controlling for experience, English level, job title, company type, employment status, programming language, and year.
 H_1 : At least one region has a statistically significant effect on salary even when all other factors are held constant.
- $H_0: \beta_{\text{region}_1} = \beta_{\text{region}_2} = \dots = 0$
 $H_1: \exists \beta_{\text{region}_j} \neq 0$

7 Results

To test our hypotheses, we used an Ordinary Least Squares (OLS) regression model using a log-transformed salary as the dependent variable. This transformation helps to normalize the distribution and interpret coefficients in percentage terms. The model included categorical variables (title, programming language, year) as fixed effects.

1. Effect of company type

- *Hypothesis:* Product companies pay higher salaries than outsourcing companies.
- *Result:*

P-value: 3.845e-51

Conclusion: We reject H_0 . Product companies statistically significantly pay more. Product companies pay 6.95% more than outsource companies on average.

- *Visualization:* Figure 6

2. Impact of English proficiency:

- *Hypothesis:* Higher levels of English proficiency are associated with higher average salaries.
- *Result:*

P-value: 0.0

Conclusion: There is an impact - as the level of English increases, the salary increases.

With the increase in English by 1 level, the average salary increases by $\approx 11.64\%$

- *Visualization: Figure 2*

3. Impact of experience:

- *Hypothesis:* Greater total experience leads to increased salary
- *Result:*

P-value: 0.0

Conclusion: Total experience increases salaries statistically significantly.

With every year of experience, the salary increases by ≈ 3.55

- *Visualization: Figure 5*

4. Effect of programming language:

- *Hypothesis:* Programming language significantly affects average salary.
- *Result:*

P-value: 5.138e-265

Conclusion: We reject H_0 , since programming language has a significant effect on wages.

- *Visualization: Figure 7*

5. Gender pay gap:

- *Hypothesis:* Gender has an effect on salary
- *Result:*

P-value: 1.865e-31

Conclusion: We reject H_0 , since gender has a statistically significant effect on wages. On average, men earn 7.29% more when other factors are fixed.

- *Visualization: Figure 3*

6. Regional salary difference:

- *Hypothesis:* Region has a significant effect on salary, controlling for other factors.
- *Result:*

P-value: 2.154e-159

Conclusion: Reject H_0 , regions have a statistically significant effect on salary.

- *Visualization: Figure 8*

7.1 Log Salary Predictions Across Regions

To further interpret the results of the hierarchical linear model, we produce predicted log salaries across different regions of Ukraine for various years. This allows us to visualize how salary levels differ across regions and how they evolve over time, accounting for both fixed and random effects in the model (Figure 9). Predicted log salaries are calculated based on the following formula:

$$\text{Predicted Log Salary}_{ij} = \beta_0 + u_{0j} + \sum_p \beta_p X_{ijp} + u_{1j} \cdot \text{Year}_{ij} \quad (4)$$

where:

- β_0 is the overall intercept (fixed effect),
- u_{0j} is the random intercept specific to region,
- β_p are the fixed effect coefficients for predictors X_{ijp} such as experience, English level, gender, full-time status, job title, and programming language,
- u_{1j} is the random slope coefficient for Year in region,
- Year_{ij} represents the year indicator variables for individual in region

Predictions assume average or baseline values for individual characteristics (e.g., a full-time male employee with average experience and English level), isolating regional and temporal differences.

7.2 Interpretation of the Plot

The plot in Figure 10 shows predicted mean log salaries for each region over different survey years. Each line represents a different region, illustrating:

- Baseline salary differences between regions, as captured by the random intercepts (u_{0j}),
- Different growth or decline rates in salaries across regions over time, as captured by the random slopes for Year (u_{1j}).

Regions with higher starting points on the plot correspond to areas where salaries are structurally higher even after controlling for individual characteristics. Steeper slopes indicate regions where salaries increased faster over time, while flatter or declining slopes highlight regions with stagnant or decreasing salary trends. This visualization highlights the importance of accounting for regional heterogeneity when

modeling salary dynamics, as both the level and trend of salary evolution vary significantly across different areas.

Additionally, we observe a pronounced effect of the 2022 Russian invasion on salaries, with regions directly affected by warfare experiencing a greater decline, whereas individuals outside of Ukraine were less impacted.

8 Conclusions

8.1 Level-Level Model

In our initial Ordinary Least Squares (OLS) model using raw salary values, we observed consistent and intuitive trends, though the model was affected by right-skewness in the dependent variable. Despite this, key independent variables, including company type, experience, English proficiency, and gender were already showing significant contributions to salary determination. However,

model statistics suggested that the residuals deviated from normality and showed signs of heteroskedasticity, therefore we decided to use a log-level approach and robust standard errors.

8.2 Log-Level Model

The final model specification applied a log transformation to the dependent variable (monthly salary), improving normality and allowing for percentage-based interpretation of coefficients. This model demonstrated strong statistical significance across all tested variables and provided a better overall fit, indicated by a lower AIC.

Below is the interpretation of the estimated coefficients:

- **Company Type:** IT professionals working in product companies earn approximately **6.95%** more than those in outsourcing firms, on average, *ceteris paribus*.
- **English Proficiency:** For each increase in English level, salary increases by approximately **11.64%** on average, *c.p.*
- **Total Experience:** Each additional year of experience increases salary by about **3.55%** on average, *c.p.*
- **Programming Language:** There are statistically significant salary differences across programming languages, with some languages commanding considerably higher wages.
- **Gender:** Male IT professionals earn **7.29%** more than female professionals, on average, *c.p.*
- **Region:** The region continues to have a statistically significant effect on salary, even when accounting for remote work and controlling for all other variables.

7.3 Hierarchical Linear Model (HLM)

The use of the Hierarchical Linear Model allowed us to effectively account for unobserved regional differences in IT salaries across Ukraine. By incorporating random intercepts and slopes at the regional level, the model captured both baseline salary variations and different growth trajectories over time among regions. This approach revealed significant regional disparities in income levels even after controlling for individual characteristics such as experience, English proficiency, and specialization, highlighting a difference in job opportunities and their development across Ukraine. Moreover, the model demonstrated that the 2022 Russian invasion had a measurable negative impact on salaries in the most affected areas. Overall, HLM improved the accuracy of our analysis, highlighting the necessity of considering geographic heterogeneity when modeling salary dynamics in the Ukrainian IT sector.

7.4 Limitations and further research

While our models provide valuable insights, they cannot capture all factors affecting IT salaries. Future research could explore additional factors such as:

- The impact of remote work on regional salary differences
- How specific industries served by IT professionals influence compensation
- The effect of specialized certifications beyond formal education
- The effect of higher education/PhD
- The influence of international market conditions on Ukraine's IT sector

8 Appendix

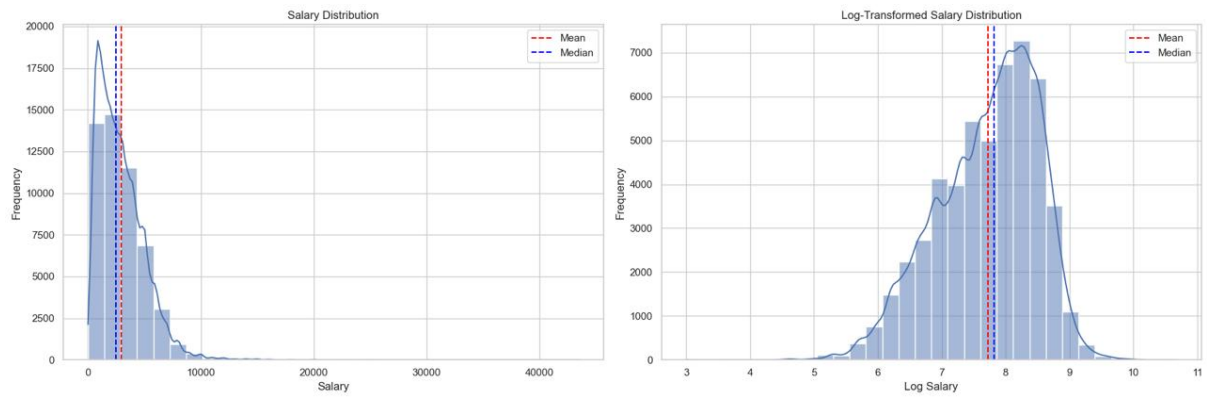


Figure 1: Distribution of Salary and $\log(\text{Salary})$

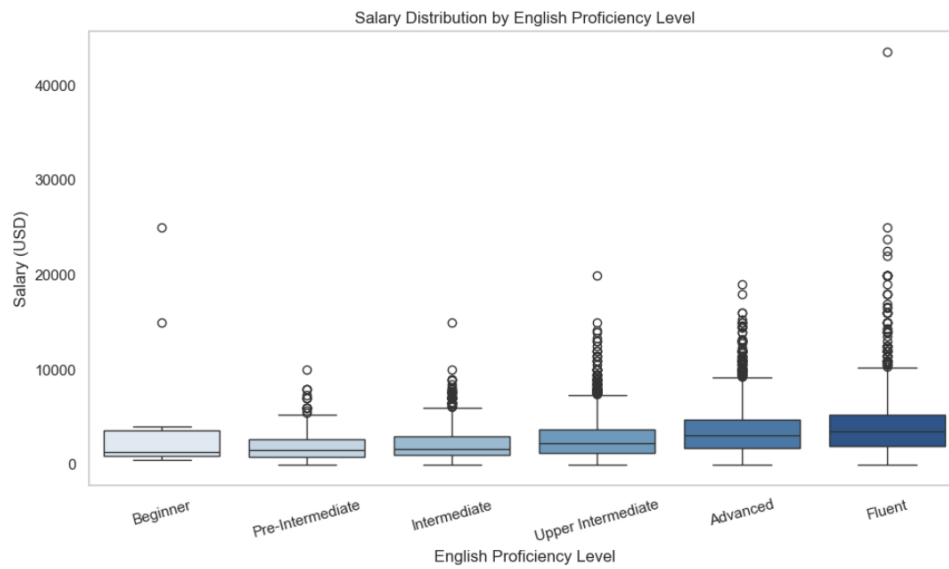


Figure 2: Salary distribution by English proficiency level

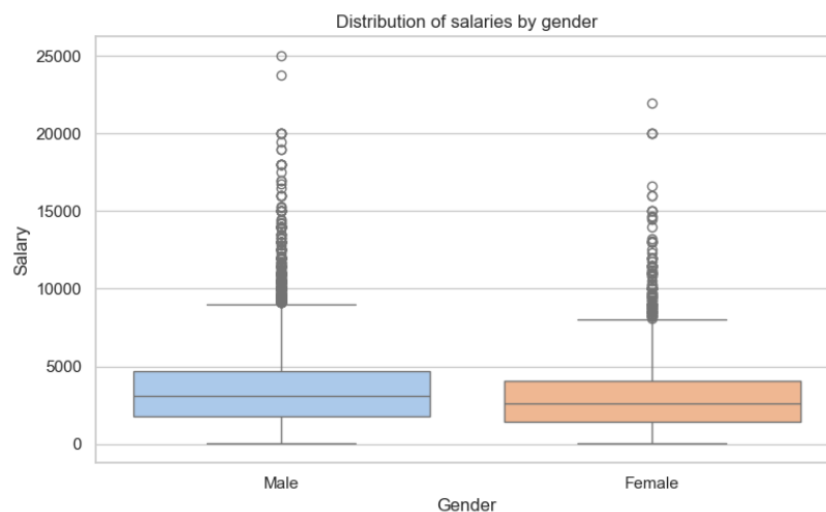


Figure 3: Distribution of salaries by gender

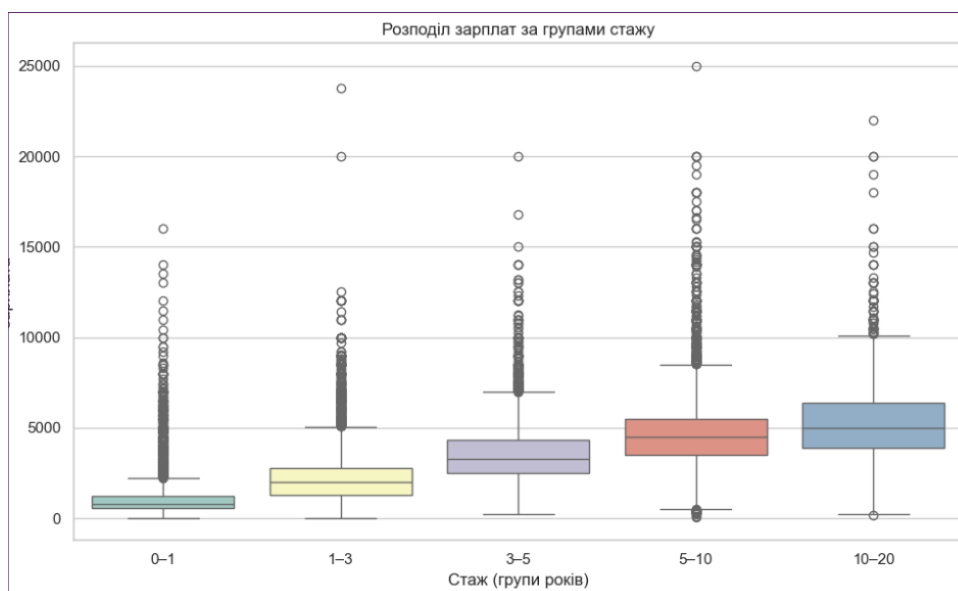


Figure 4: Salary distribution by total experience

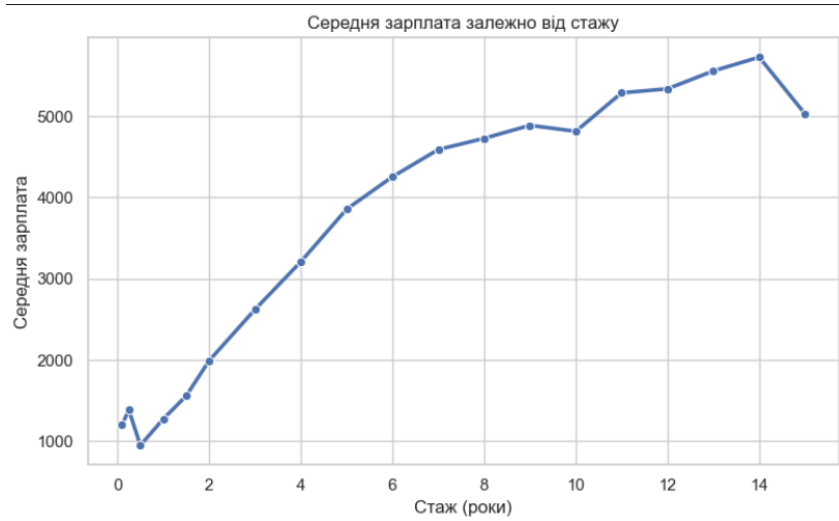


Figure 5: Effect of total experience on Salary



Figure 6: Average Salary difference between Non-product and Product companies



Figure 7: Distribution of salary in top 10 programming languages

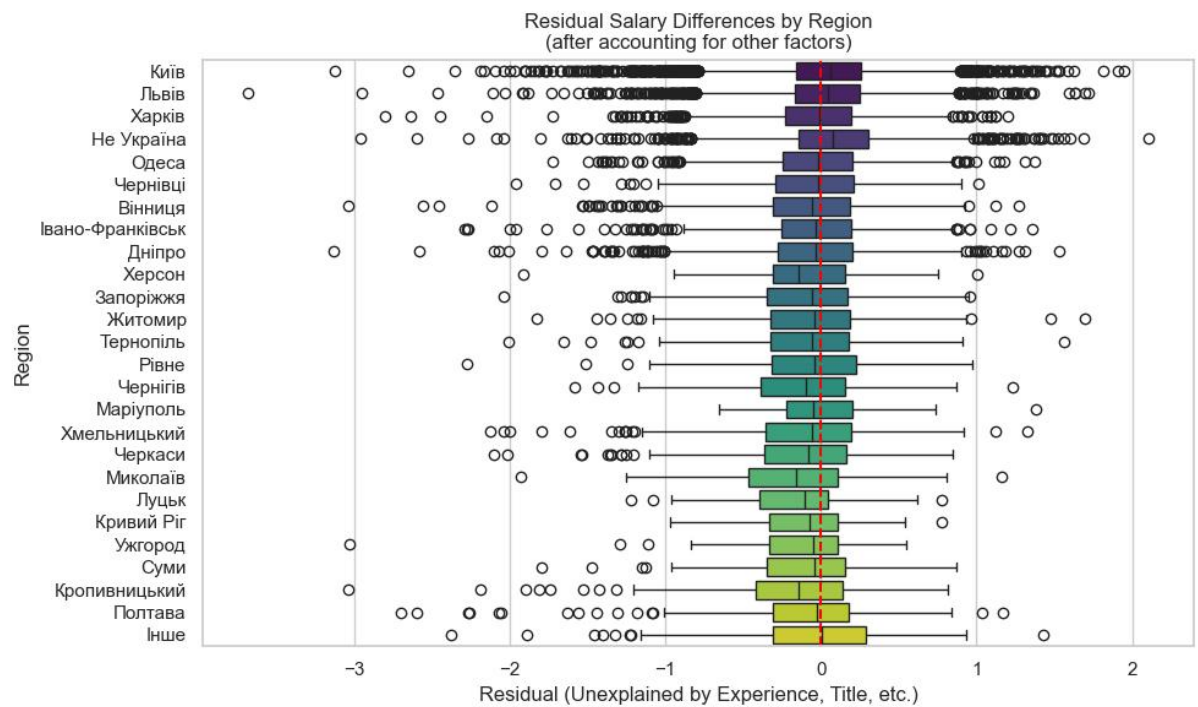


Figure 8: Average salary by region

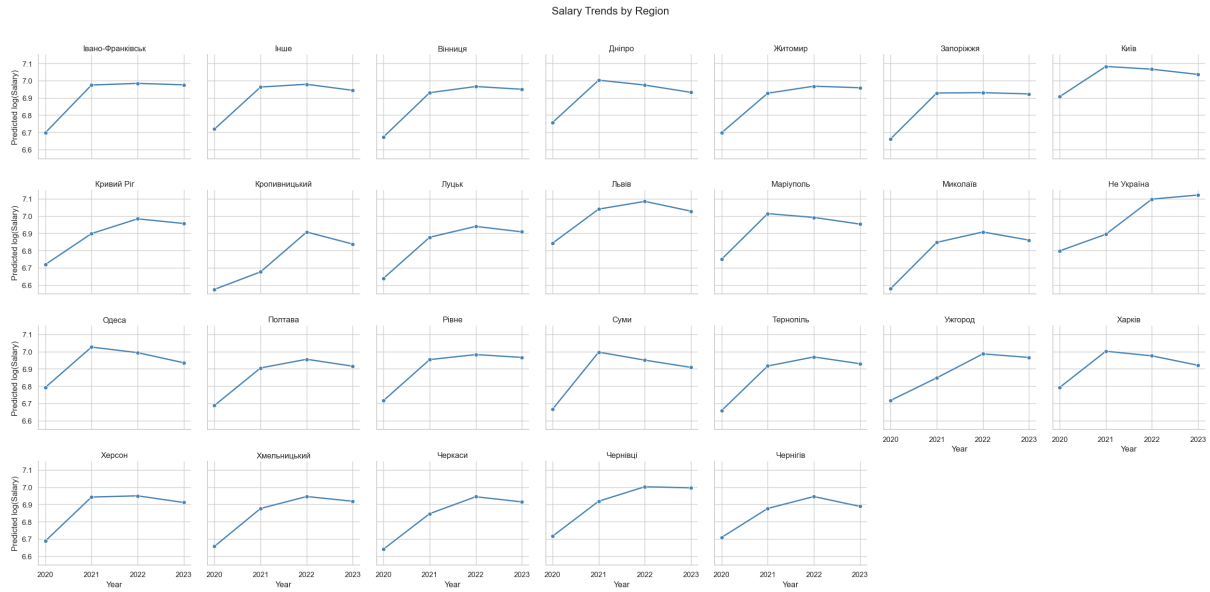


Figure 9: Predicted $\log(\text{Salary})$ trend for each region

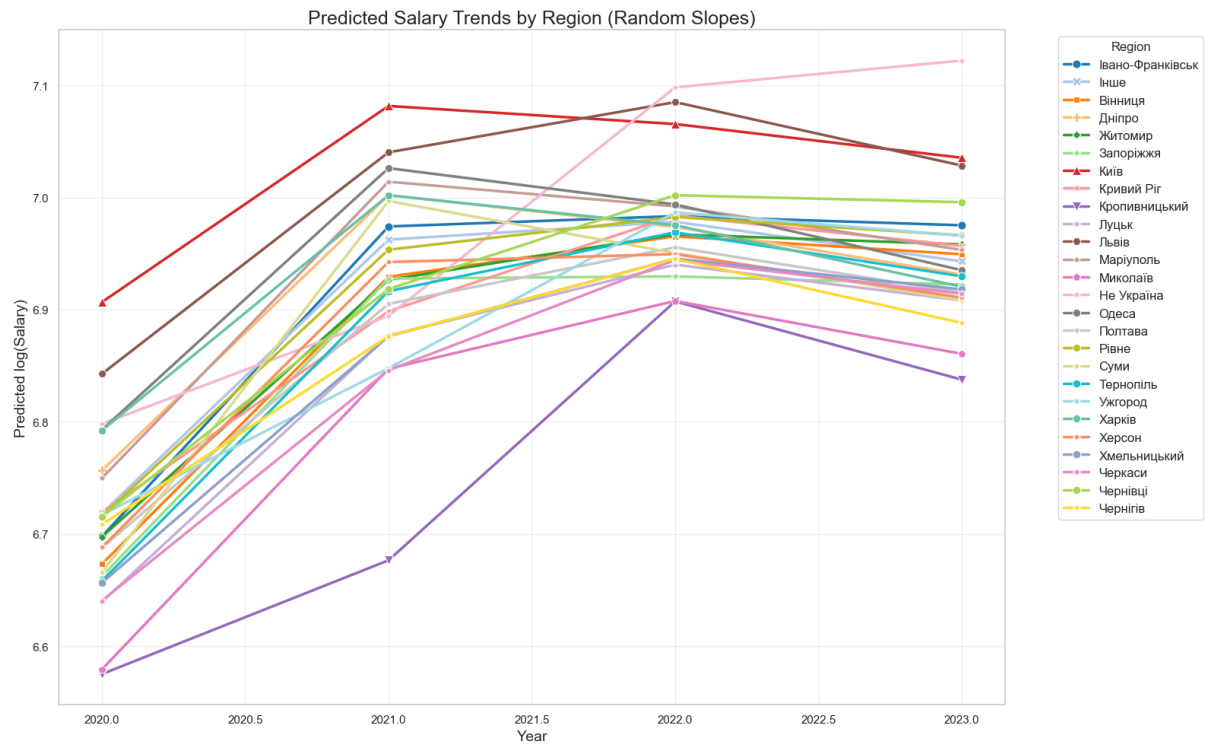


Figure 10: Combined $\log(\text{Salary})$ trends for all regions

Code The code used in project research is available on GitHub.